



Department of Computer Science & Engineering Uttara University

MIS0611403 - System Auditing & Maintenance

Conducted By

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WELCOME

Course Information

- Course Code : MIS0611403
- Course Title: System auditing and maintenance
- Course Credit : 3.00

Course Rationale

- Enhances understanding of Information Systems auditing & maintenance
- Covers evaluation of security and controls
- Includes techniques for external EDP audits
- Introduces CAATs (Computer-Assisted Audit Techniques)
- Addresses auditing of EDI systems
- Examines audit & control issues in eCommerce, networks, VPNs, and continuous auditing
- Explores audit approaches and computer-related risks
- Focuses on internal controls and their concerns
- Covers evaluation of systems and business processes
- Develops troubleshooting and problem-solving skills for user and system issues

Course Learning Outcome

CLO1	Describe the systematic process of information security audit.
CLO2	Demonstrate an independent professional audit activity to support the incident investigation.
CLO3	Analyze the security incident with a proven audit procedure.
CLO4	Generate an audit reporting with findings.
CLO5	Analyze common symptoms and problems associated with system and provide solutions.

Grading Scale

Numerical Grade	Letter Grade		Grade Points	Comment
80% and above	A+	A (plus)	4.00	Outstanding
75% to Less than 80%	A	A (regular)	3.75	Excellent
70% to Less than 75%	A-	A (minus)	3.50	Very good
65% to Less than 70%	B+	B (plus)	3.25	Good
60% to Less than 65%	B	B (regular)	3.00	Above average
55% to Less than 60%	B-	B (minus)	2.75	Average
50% to Less than 55%	C+	C (plus)	2.50	Below average
45% to Less than 50%	C	C (regular).	2.25	Fair
40% to Less than 45%	D		2.00	Pass
Less than 40%	F		0.00	Fail
	Irregular grading status: F : Failure I : Incomplete W : Withdrawal R : Retake			

Outline

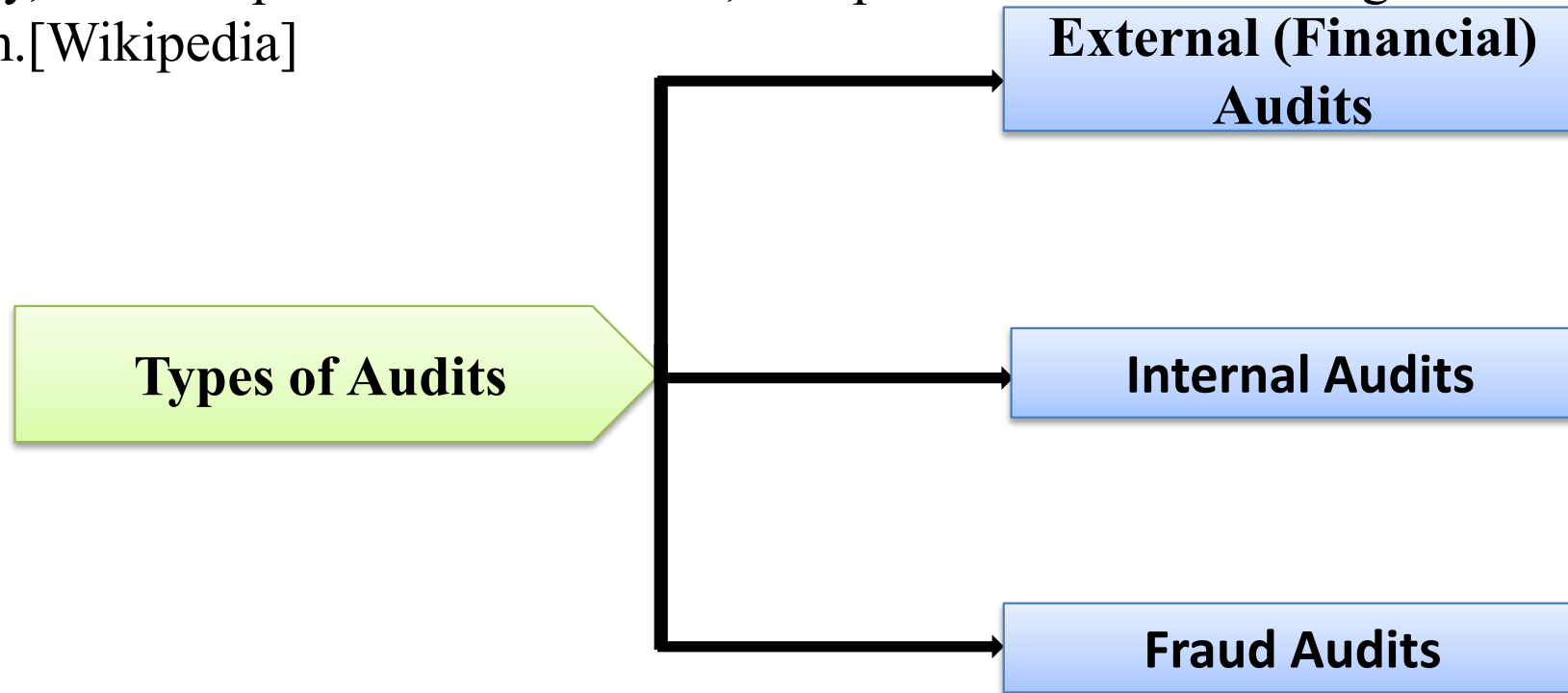
- ❑ Why MIS0611403?
- ❑ Introduction to Information Systems Auditing
- ❑ Evolution of auditing in IT environments
- ❑ Role of auditors in modern organizations
- ❑ Overview of risks, controls, and assurance

Why MIS0611403?

- Information Systems auditing ensures secure, reliable, and trustworthy business operations
- Organizations must protect systems from fraud, errors, and unauthorized access
- Modern audits use EDP techniques and CAATs to analyze large volumes of digital data
- System auditing helps identify, assess, and mitigate computer-related business risks
- The course builds troubleshooting and problem-solving skills for real system issues

OVERVIEW OF AUDITING

Audit: An audit is **an independent examination of financial information** of any entity, whether profit oriented or not, irrespective of its size or legal form.[Wikipedia]



External versus Internal Auditors

- External auditors represent outsiders while internal auditors represent the organization
- Internal auditors assist external auditors to increase efficiency and reduce audit fees
- Internal auditors reporting to the controller lack independence—external auditors cannot rely on their work
- Independent internal auditors can gather evidence year-round, making external audits more efficient and less costly

What is Information Systems Auditing & Why We need it?

IS Auditing

- Information Systems Auditing focuses on reviewing computerized systems to ensure data accuracy, system security, and proper control mechanisms. It goes beyond financial auditing

Importance

- Almost all business activities are now technology-driven. If systems fail or are manipulated, business decisions become unreliable, making auditing essential

Evolution of Auditing

- Earlier audits focused on paper-based records. With the introduction of computers, auditors had to adapt their techniques to evaluate electronic data and automated controls.

Traditional Auditing vs IT Auditing

- Traditional auditing focuses on outputs, while IT auditing evaluates the *systems*, *processes*, and *controls* that generate those outputs.

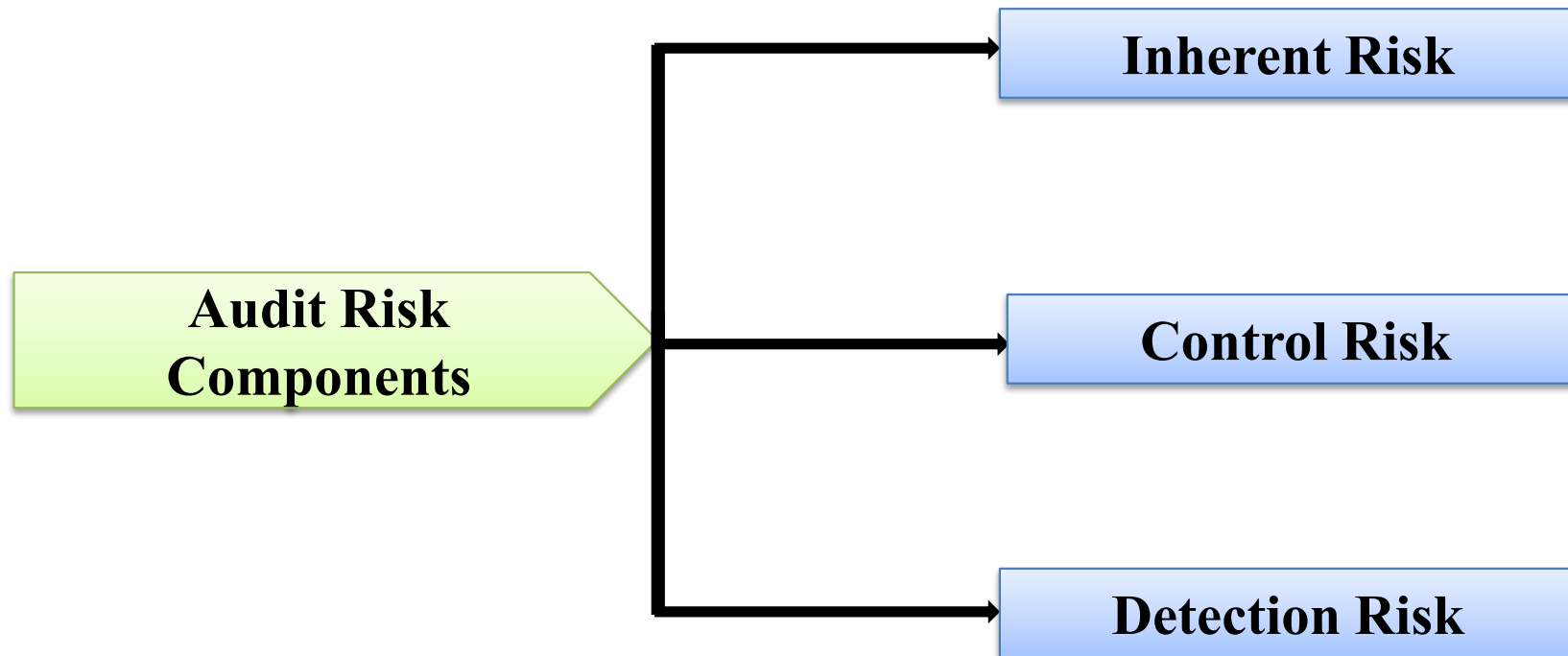
Key Objectives of IS Auditing

The **main objectives** of IS auditing are

- Protecting assets
- Maintaining data accuracy
- Ensuring system availability
- Supporting organizational goals

AUDIT RISK

Audit risk is the probability that the auditor will render an unqualified (clean) opinion on financial statements that are, in fact, materially misstated



AUDIT RISK MODEL

Audit Risk Model

Financial auditors use the audit risk components in a model to determine the scope, nature, and timing of substantive tests. The audit risk model is

$$AR = IR \times CR \times DR$$

Assume that acceptable audit risk is assessed at a value of 5 percent, consistent with the 95 percent confidence interval associated with statistics. By illustration, assume IR is assessed at 40 percent, and CR is assessed at 60 percent. What would be the level of planned detection risk (DR) needed to achieve the acceptable audit risk (AR) of 5 percent?

$$5\% = 40\% \times 60\% \times DR$$

$$DR = .05 / .24$$

$$DR = .20$$

Let's now reduce the control risk (CR) value to 40 percent and recalculate DR.

$$5\% = 40\% \times 40\% \times DR$$

$$DR = .31$$

The Structure of an IT Audit

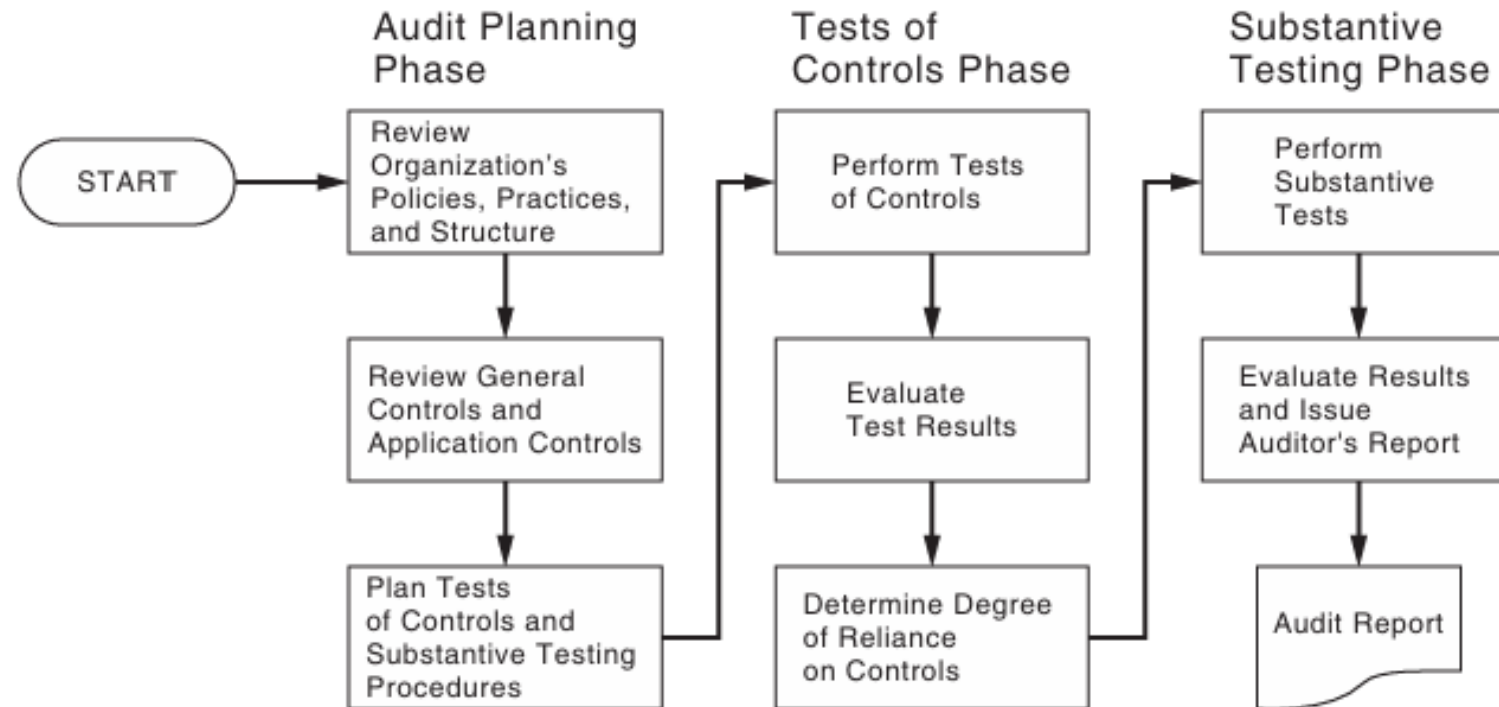


Fig 1: The Structure of an IT Audit [1]

Internal Control

Internal control is a process effected by an entity's board of directors, management, and other personnel designed to provide reasonable assurance regarding the achievement of objectives in three categories: effectiveness and efficiency of operations, reliability of financial reporting, and compliance with applicable laws and regulations

INTERNAL CONTROL OBJECTIVES

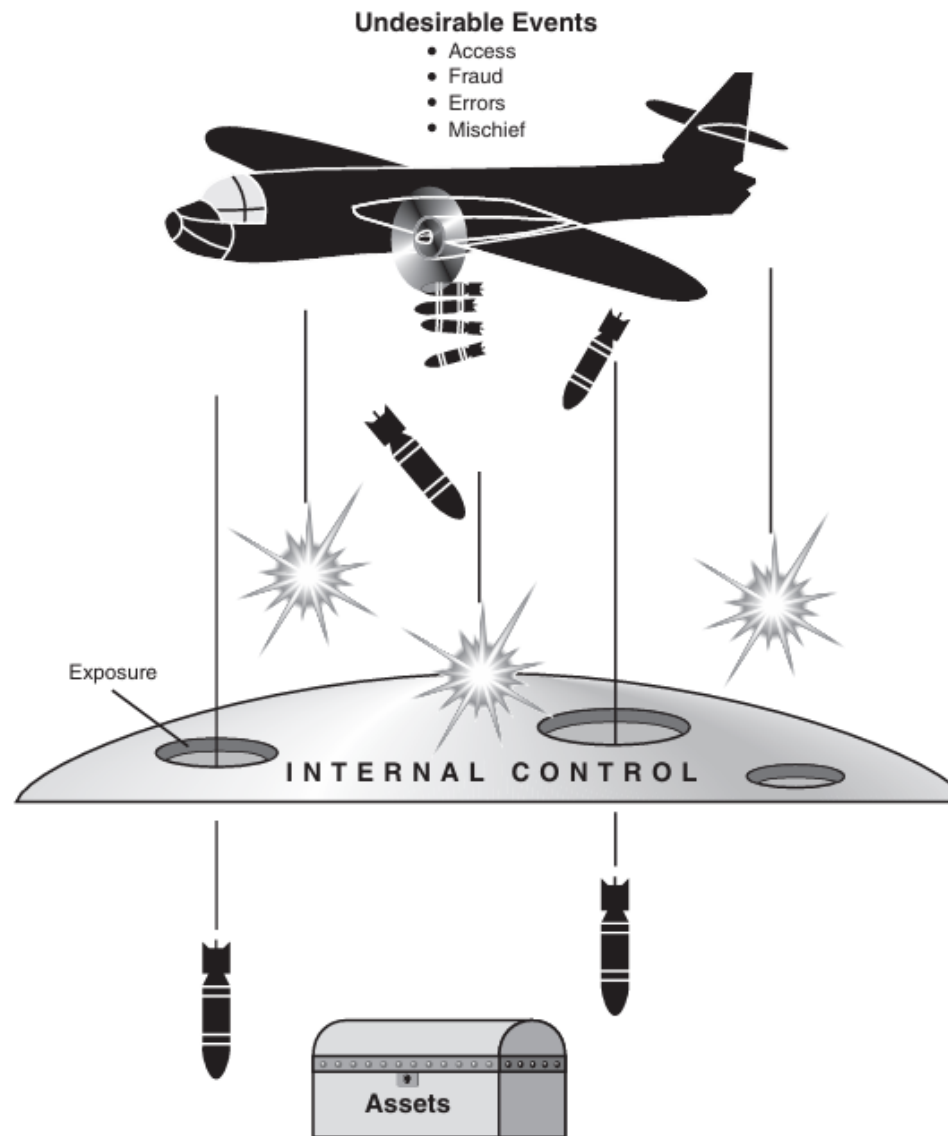
An organization's **internal control system** comprises policies, practices, and procedures to achieve four broad objectives:

1. To safeguard assets of the firm.
2. To ensure the accuracy and reliability of accounting records and information.
3. To promote efficiency in the firm's operations.
4. To measure compliance with management's prescribed policies and procedures.⁷

INTERNAL CONTROL OBJECTIVES

FIGURE 1.2

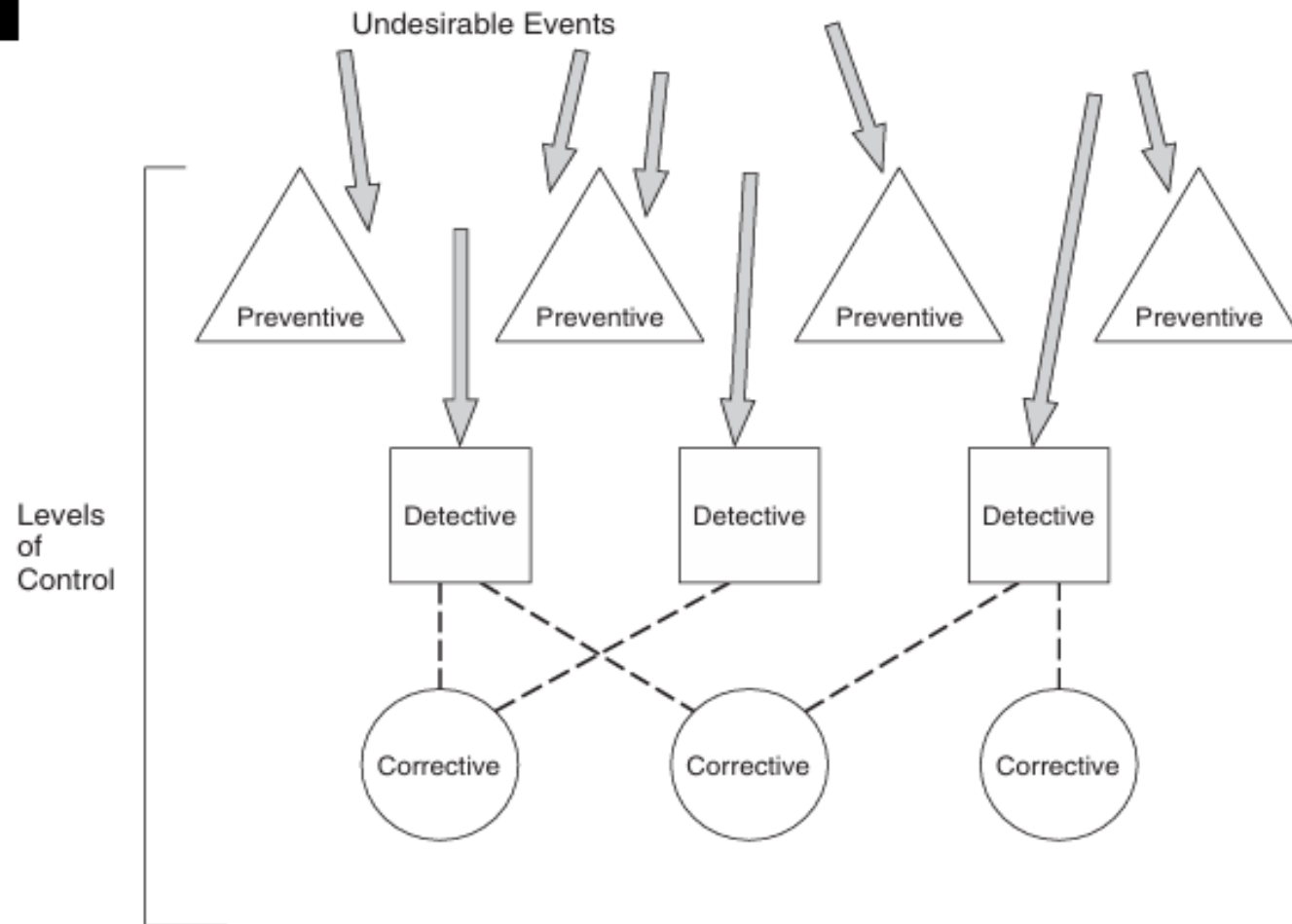
**Internal Control
Shield**



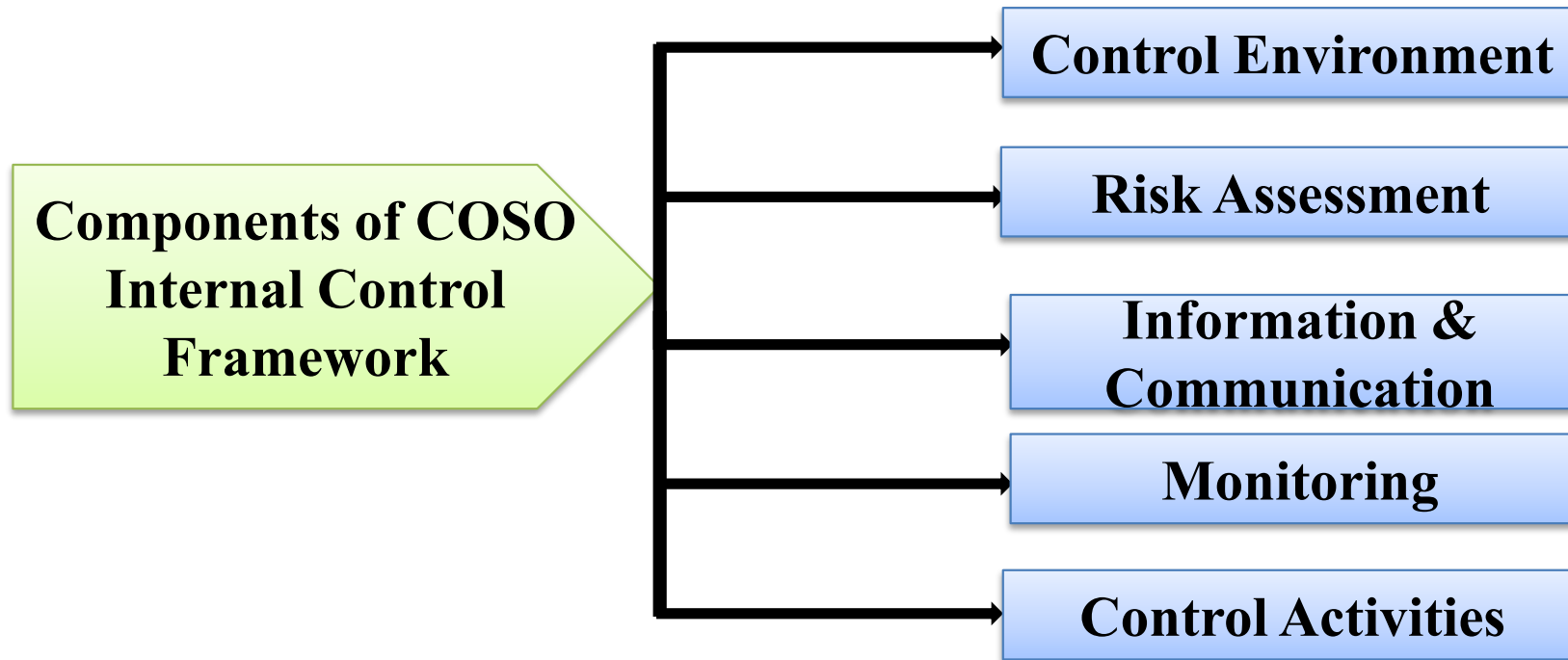
The PDC Control Model

FIGURE 1.3

**Preventive,
Detective, and
Corrective
Controls**



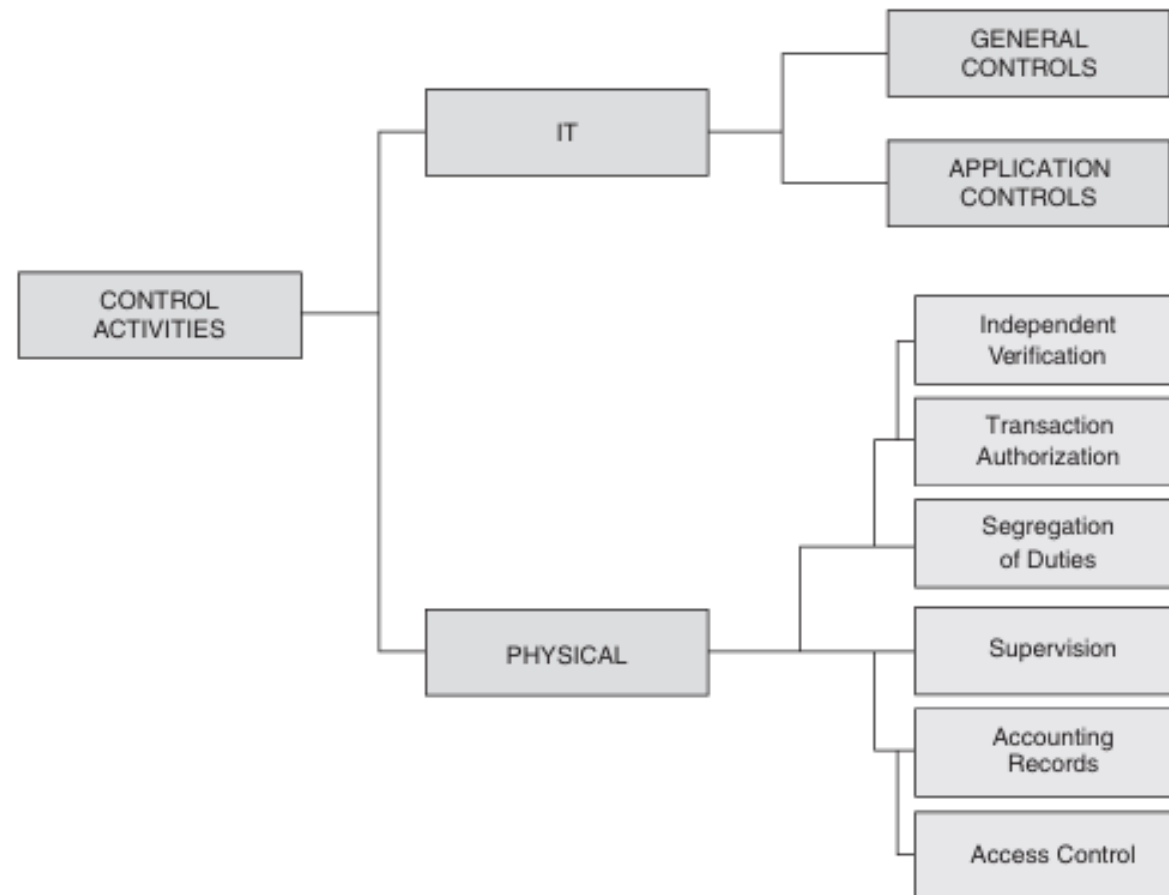
COSO Internal Control Framework



Categories of Control Activities

FIGURE 1.4

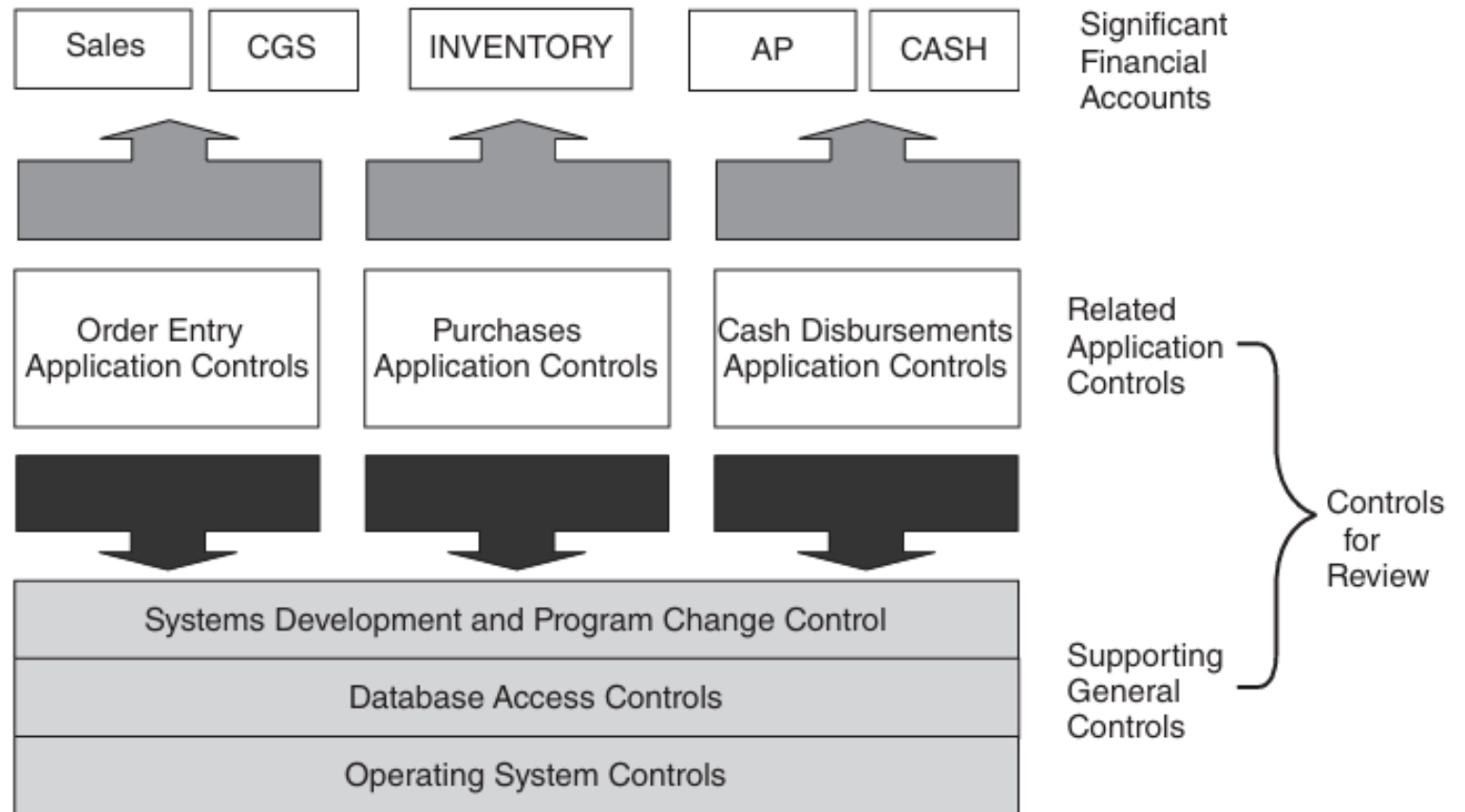
**Categories of
Control
Activities**



IT Control Relationship

FIGURE 1.6

**Information
Technology
Control
Relationship**



Reference

[1] James A. Hall: *Information Technology Auditing & Assurance*

THANK YOU